



Dense regular

Dense irregular

Adipose

Loose Areolar

Reticular

Elastic



Hochschule Aalen

Histology of Soft Connective Tissues

Thomas F. Freddo, O.D., Ph.D.

Professor

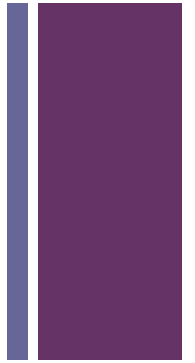
MCP Health Sciences University

+ Connective tissues serve several important roles

- Soft connective tissues are the main passageways beneath epithelium and within fascicles of muscle tissues and large nerves that provide the requisite continuous passageway through which nerves and vessels travel in order to reach their target tissues.
- Loose connective tissues also serve as the most common sites for battles fought by the immune system against foreign antigen or infectious agents.
- It is the location where most of the critical steps in antigen recognition and presentation of antigen to the immune system occur as part of the reticuloendothelial system.
- The extracellular matrices created by the connective tissue cells can also play a role in filtering macromolecular solutes



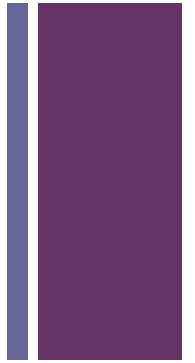
Connective Tissue



- Cells
- Extracellular Matrix
 - Structural proteins
 - Ground substance

+ Connective Tissue Cells

- Fibrocytes/Fibroblasts
- Mast cells
- Macrophages
- Leukocytes and plasma cells (B cell-derivative)
- Adipocytes

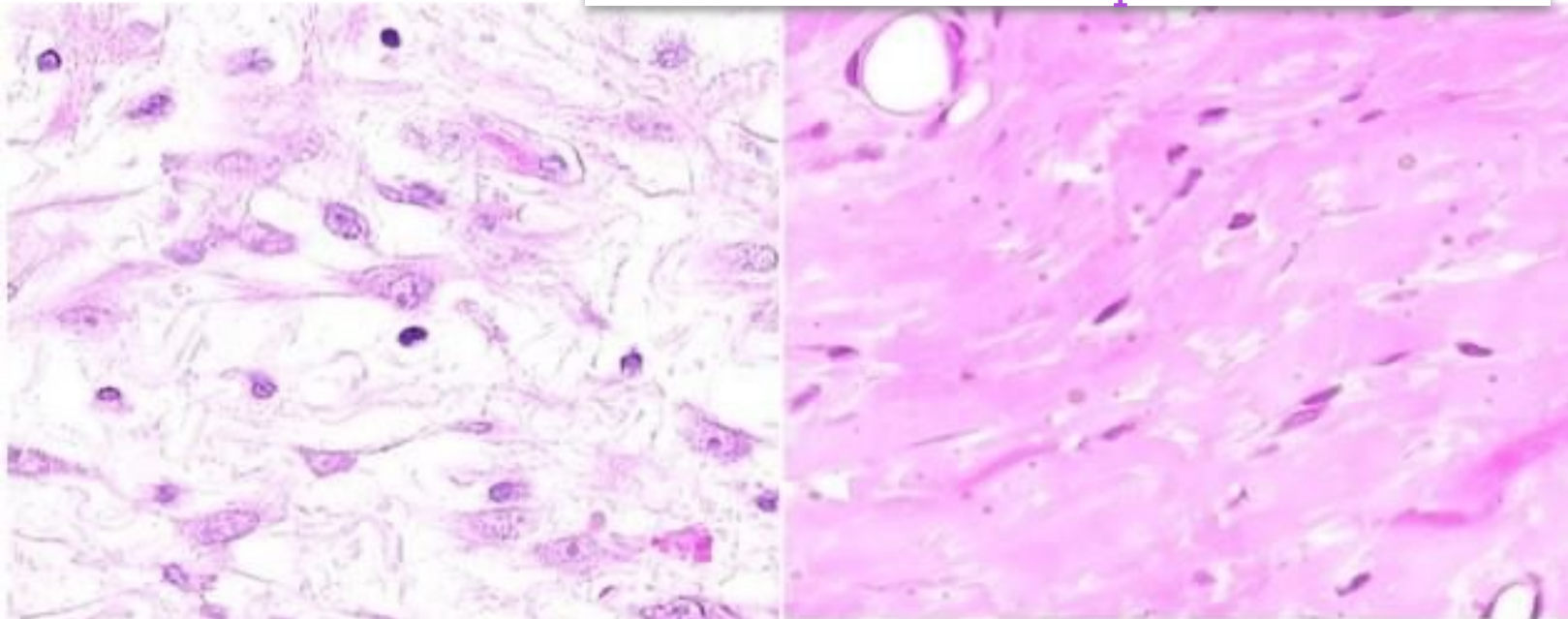


+ Connective Tissue Cells

Fibrocytes/Fibroblast

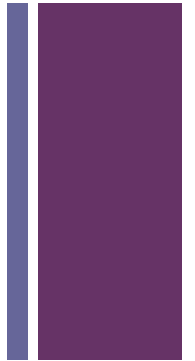


Active form = fibrocyte = more ovoid
nuclei with multiple processes
Inactive cell = fibroblast – thin fusiform
cells with few processes



+ Connective Tissue Cells

Fibrocytes/Fibroblasts

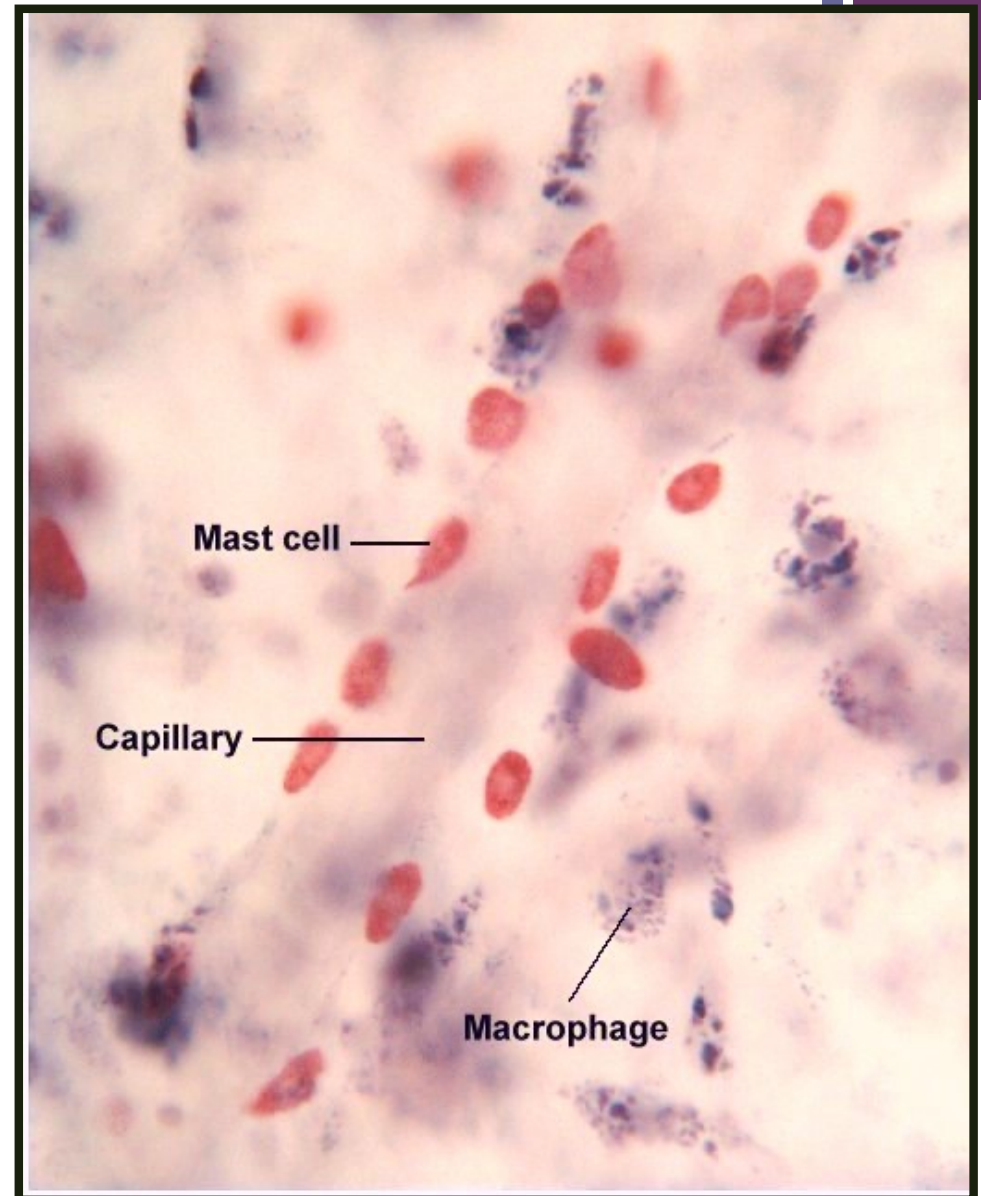


- Fibroblasts, when activated to fibrocytes make collagens, proteoglycans, glycosaminoglycans, reticular and elastic fibers, found in the extracellular matrix.
- They also induce chemokine synthesis through the presentation of receptors on their surface. Immune cells then respond and initiate a cascade of events to clear the invasive microorganisms.

+ Connective Tissue Cells

■ Mast Cells

- Mast cells are present in most tissues characteristically surrounding blood vessels and nerves, and are especially prominent near the boundaries between the outside world and the internal milieu, such as the skin, mucosa of the lungs, and digestive tract, as well as the mouth, conjunctiva, and nose



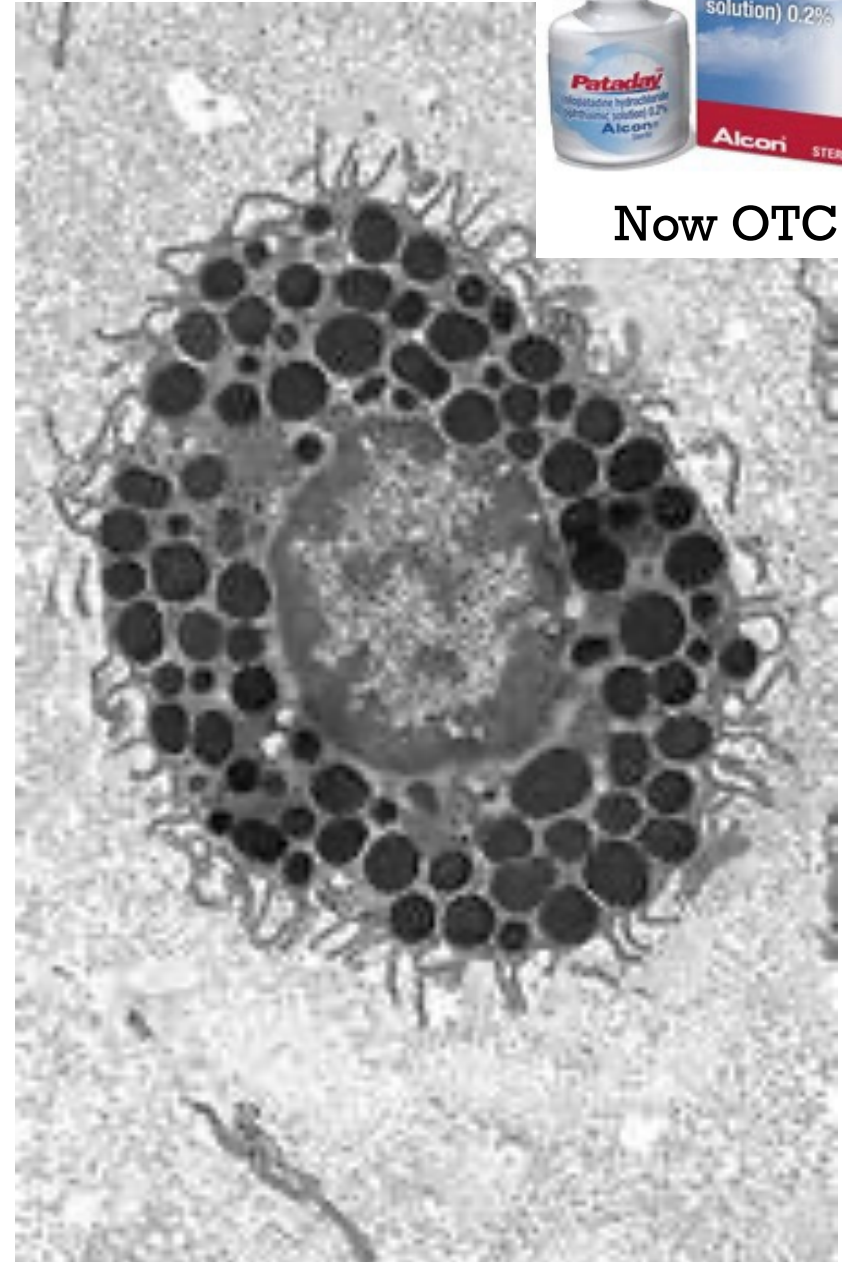
+ Connective Tissue Cells

■ Mast Cells

- Mast cells contain granules containing preformed vasoactive mediators (e.g. histamine)
- Their membranes maintain IgE receptors
- If an allergen is encountered, binding of antigen triggers “degranulation” – a process leading to fusion of the membrane-bound granules to the cell membrane and release of the granule contents.
- If the cell membrane is “stabilized”, the granules cannot fuse and their contents cannot be released.



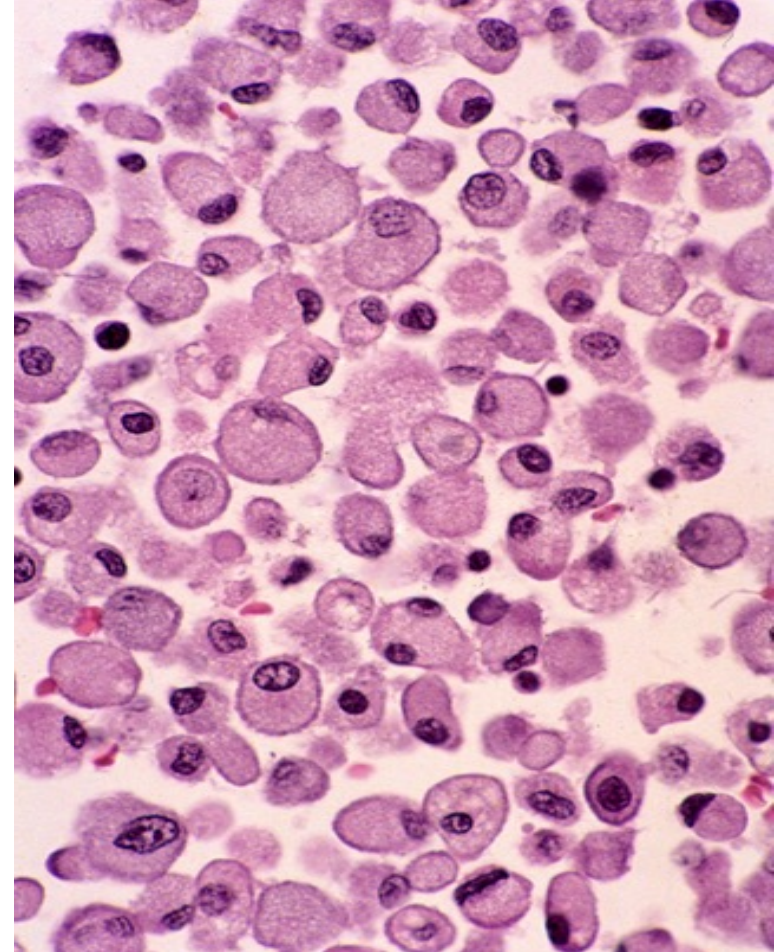
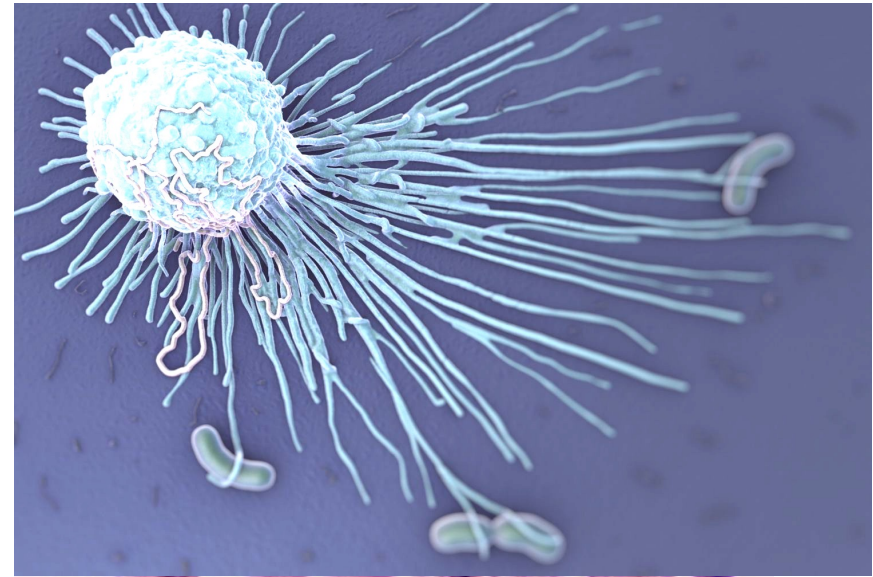
Now OTC



+Connective Tissue Cells

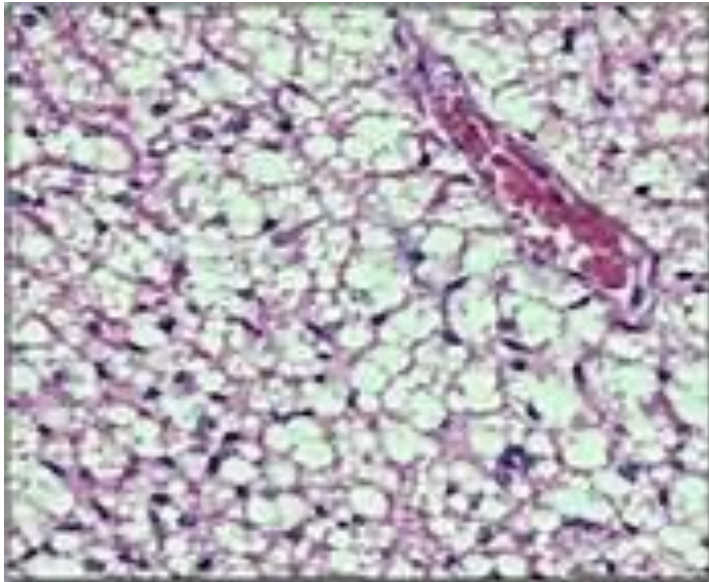
■ Macrophages- derived from monocytes

- Large phagocytes patrolling by amoeboid movement
- They take various forms (with various names) throughout the body (e.g., histiocytes, Kupffer cells, alveolar macrophages, microglia, and others), but all are part of the mononuclear phagocyte system (aka reticuloendothelial system). Besides phagocytosis, they play a critical role in immunity)
- They are important as antigen presenters to T cells in connective tissues

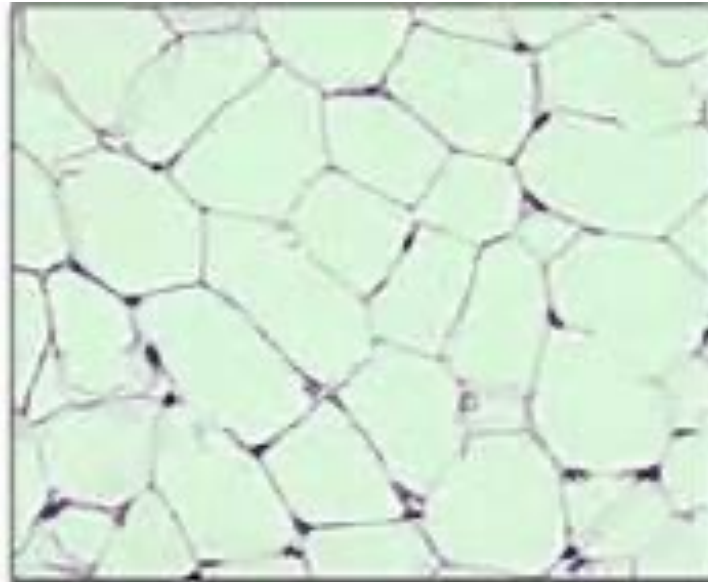


+ Adipocytes – recall that fats get washed out of cells during processing

- Cells specialized in storing fat.
- White Fat – (subcutaneous fat – hypodermis)
 - The tissue as a whole provides lipid storage, primarily as triglycerides and cholesterol esters.
 - The tissue as a whole provides cushioning to decrease tissue damage, and pain fiber stimulation
- Brown Fat – (surrounding organs)
 - This form of adipose tissue (brown fat) provides thermal insulation.

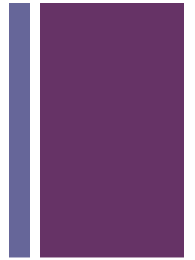


Brown adipocytes



White adipocytes

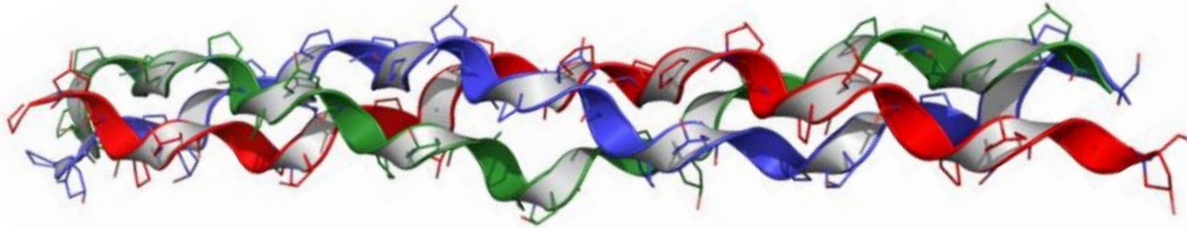
+ EXTRACELLULAR MATRIX STRUCTURAL PROTEINS



Tissue	Purpose	Components	
Collagens	Tensile strength	Alpha polypeptide helical chains	Tendons, ligaments, dermis, cornea, bone, blood vessels, intervertebral discs
Elastic fibers	Stretch and recoil	Elastin and associated microfibrils	Various
Reticular fibers	Scaffoldings for cells	Type III collagen	Liver, bone marrow and lymphatic organs

+ Collagens – almost 20 types

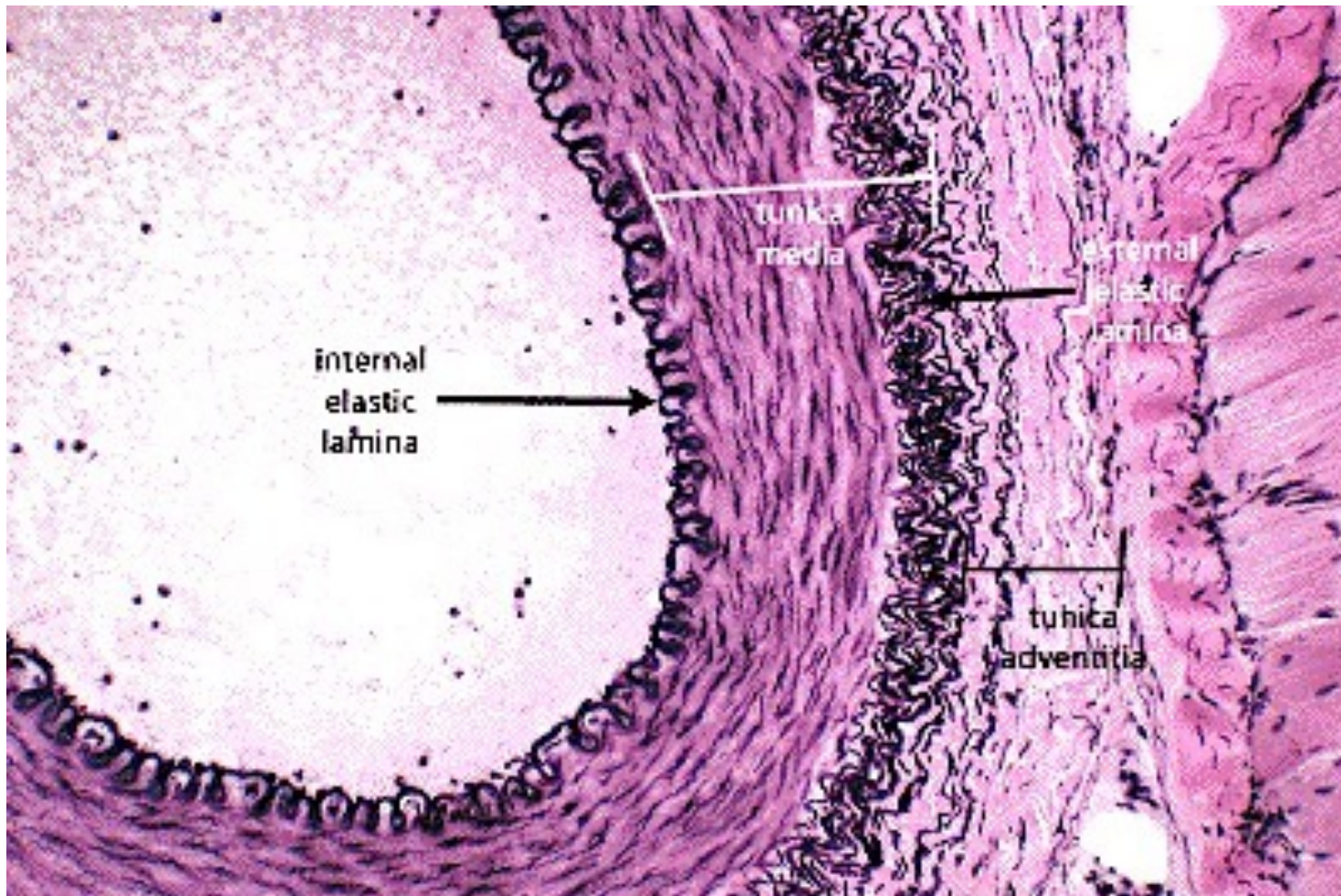
Collagens (type I collagen makes up 25% of protein mass of the body)
High tensile strength, little stretch
Alpha helical fibers (triple helix) created primarily by fibroblasts



- The five most common types are:
- Type I: skin, tendon, vascular ligature, organs, bone (main component of the organic part of bone)
- Type II: cartilage (main collagenous component of cartilage)
- Type III: reticulate (main component of reticulin fibers), commonly found alongside type I.
- Type IV: forms basal lamina, the epithelium-secreted layer of the basement membrane.
- Type V: cell surfaces, hair and placenta

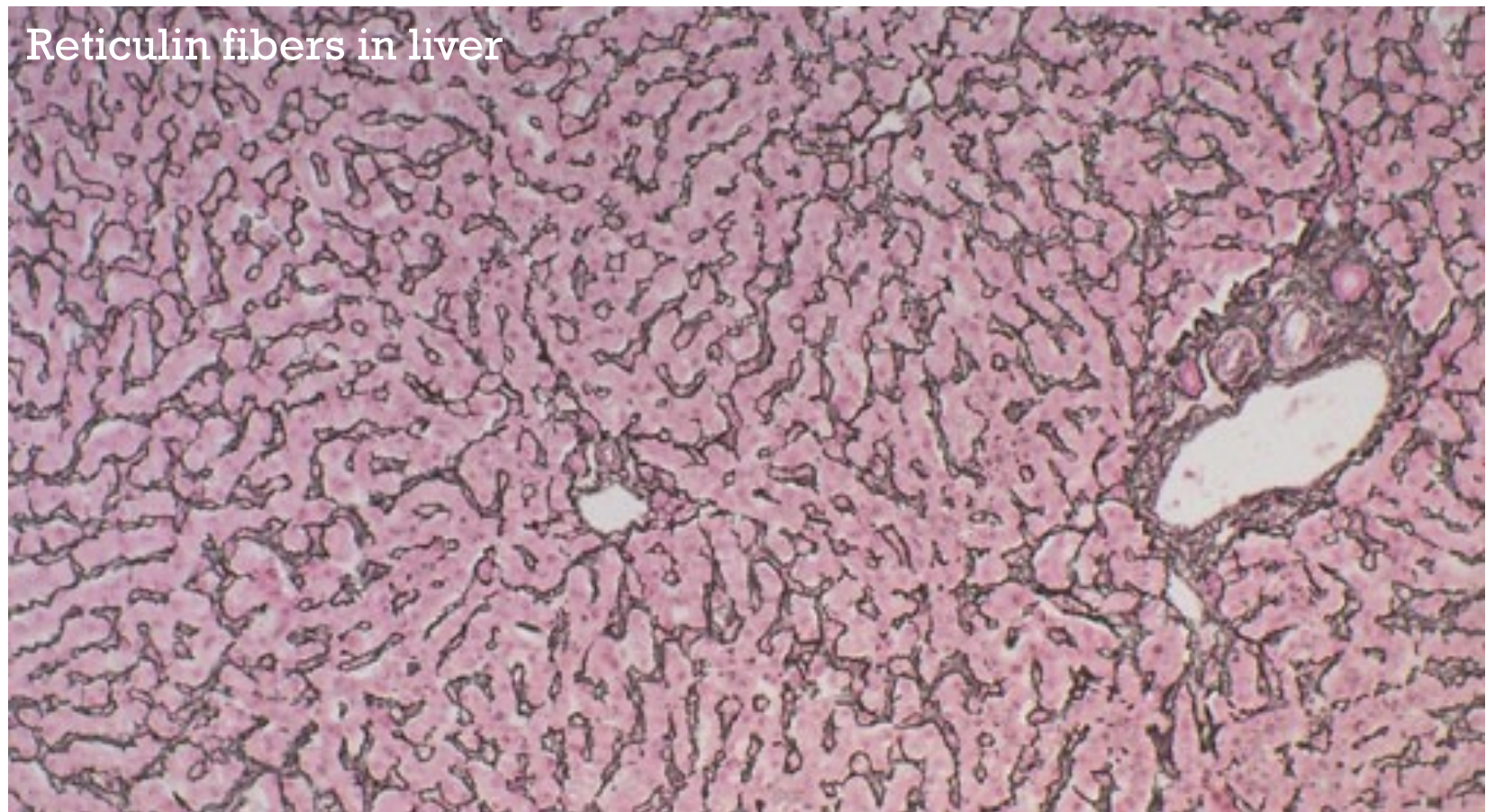
+ ELASTIC Fibers

- Fibers provide elasticity – they can stretch to 1.5 times their length and, importantly, snap back to original size.
- The fibers are composed of microfibrils (made of fibrillins) associated with amorphous elastin.

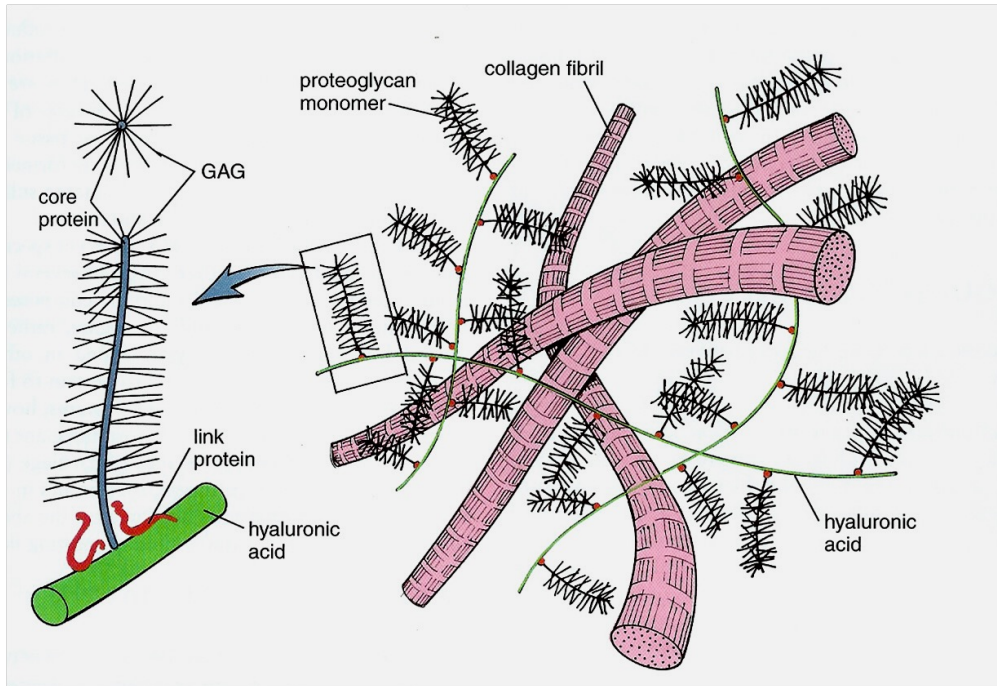


+ Reticulin (Type III Collagen)

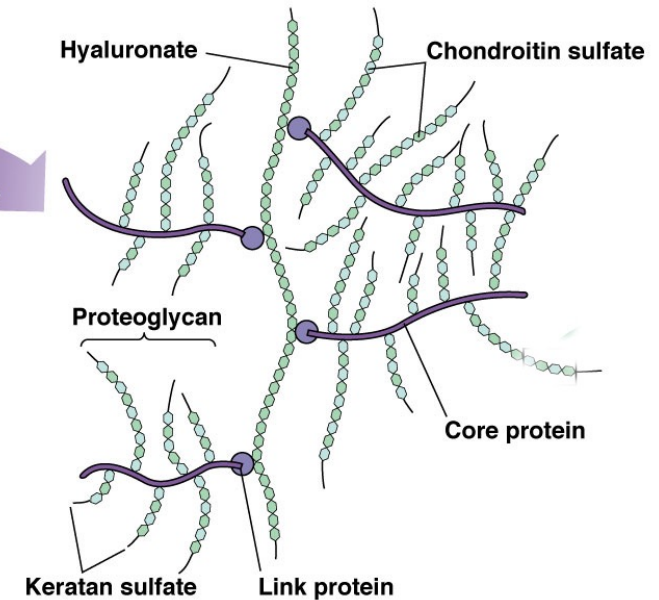
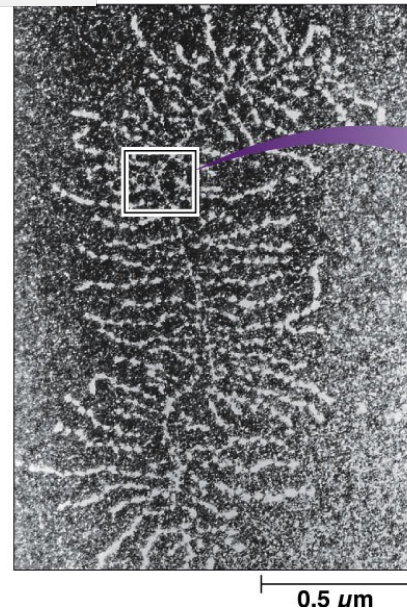
Originally defined by their unique affinity for silver based stains or PAS+. Today, we know that these fibers are composed of collagen type III.



+ GROUND SUBSTANCE



■ Relationship between Collagen, Proteoglycans, Glycosaminoglycans

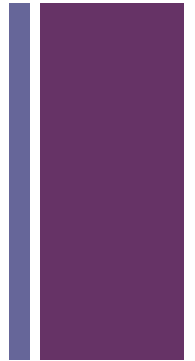


+ PROTEOGLYCANs/Glycosaminoglycans

- Proteoglycans are heavily glycosylated proteins
- There is one of several “core” proteins to which various glycosaminoglycans (formerly called mucopolysachharides) are added.

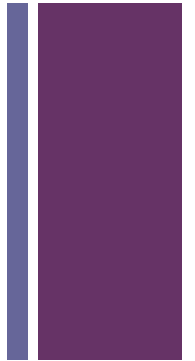
Large Proteoglycans	Small Proteoglycans	Glycosaminoglycans
versican	Decorin, biglycan	Chondroitin/dermatan sulfate
perlecan	testican	heparAn sulfate/chondroitin sulfate
neurocan aggrecan (cartilage/sclera) brevican	bikunin	Chondritin sulfate
	Fibromodulin	KeratAn sulfate

+ Non-sulfated Glycosaminoglycan HYALURONAN



- Unique among glycosaminoglycans
- It is non-sulfated
- Forms in the plasma membrane instead of the Golgi apparatus,
- Can be very large, with its molecular weight often reaching the millions.
- Binds large amounts of water and holds it in place like a sponge
- One of the chief components of the extracellular matrix (and virtually all of the matrix in the vitreous), hyaluronan contributes significantly to cell proliferation and migration, and may also be involved in the progression of some malignant tumors.
- The average 70 kg (154 lb) person has roughly 15 grams of hyaluronan in the body, one-third of which is turned over (degraded and synthesized) every day.

+ Histological Categories Based on Light Microscopic Organization



Loose

Areolar Connective Tissue

Adipose Tissue

Reticular Connective Tissue

Dense

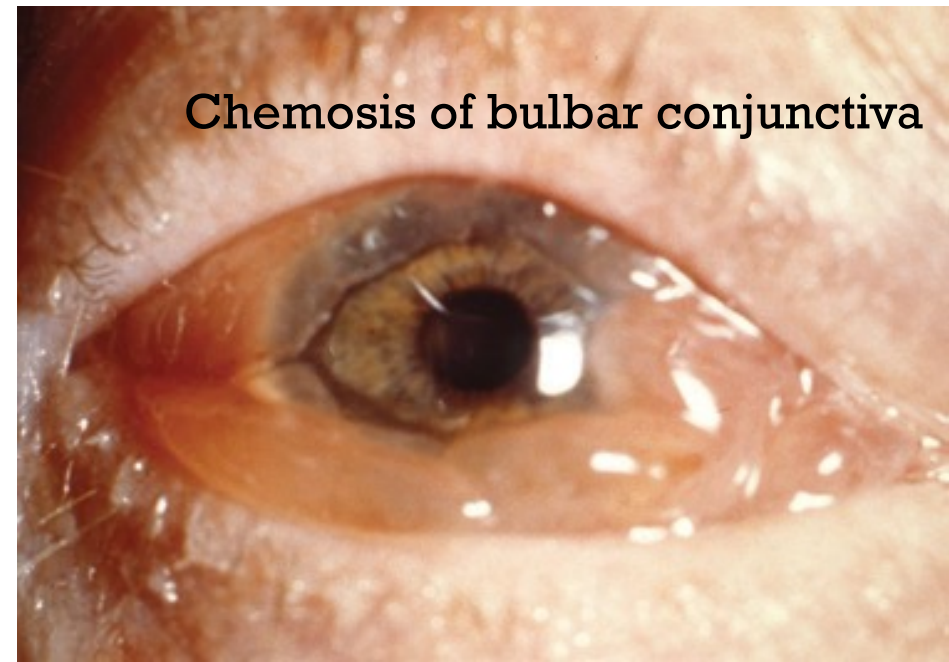
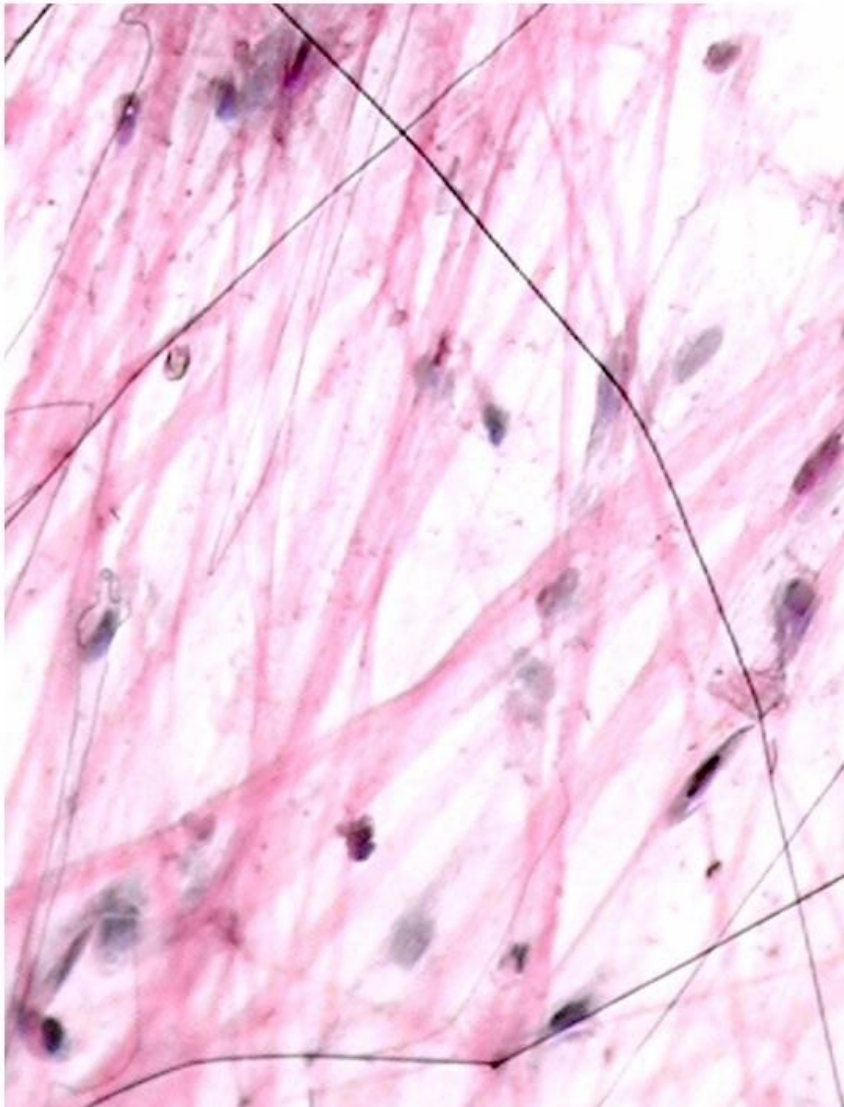
Dense Regular Connective Tissue

Dense Irregular Connective Tissue

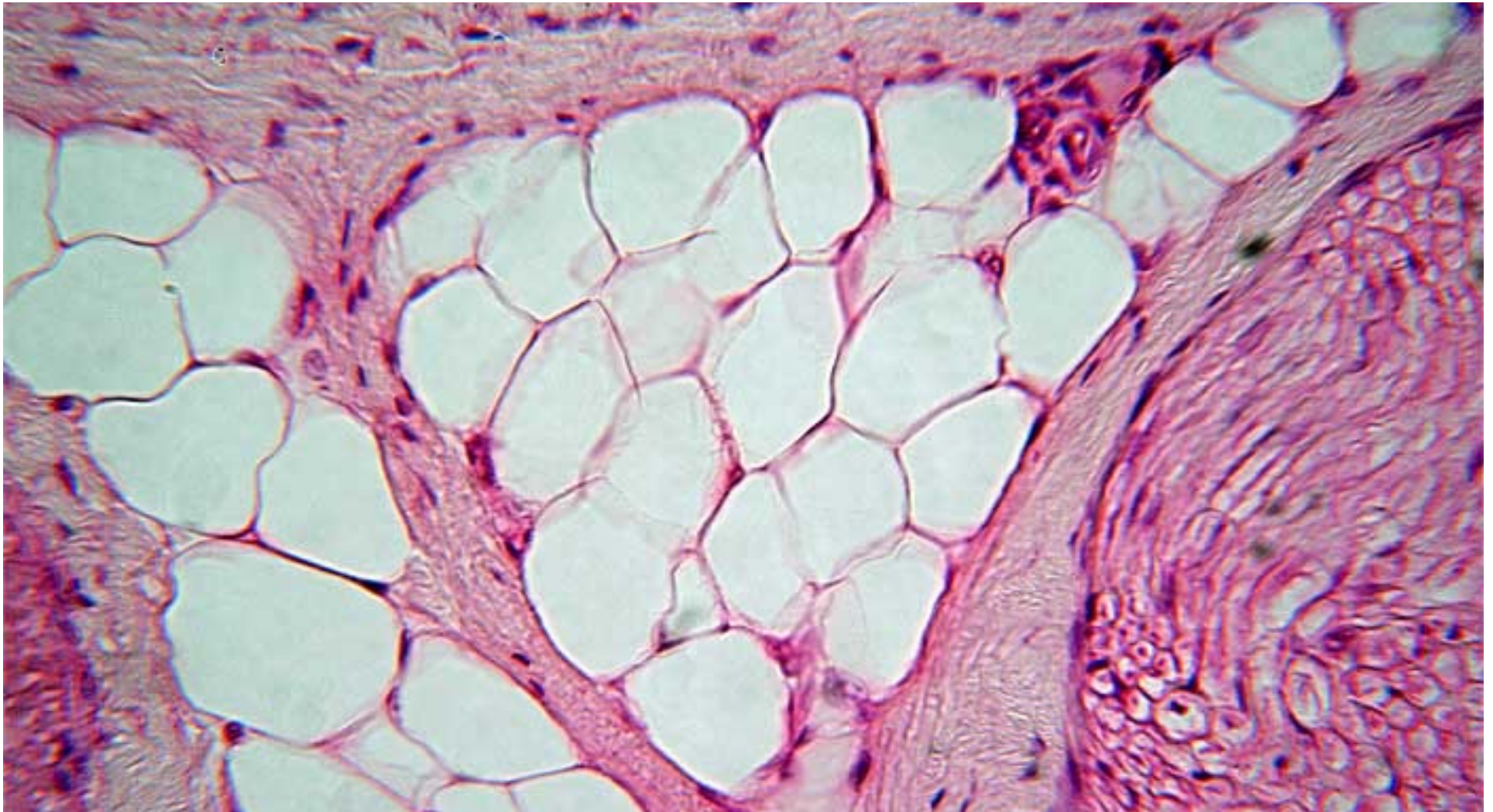
Elastic Connective Tissue

+ Loose (Areolar) Connective Tissue

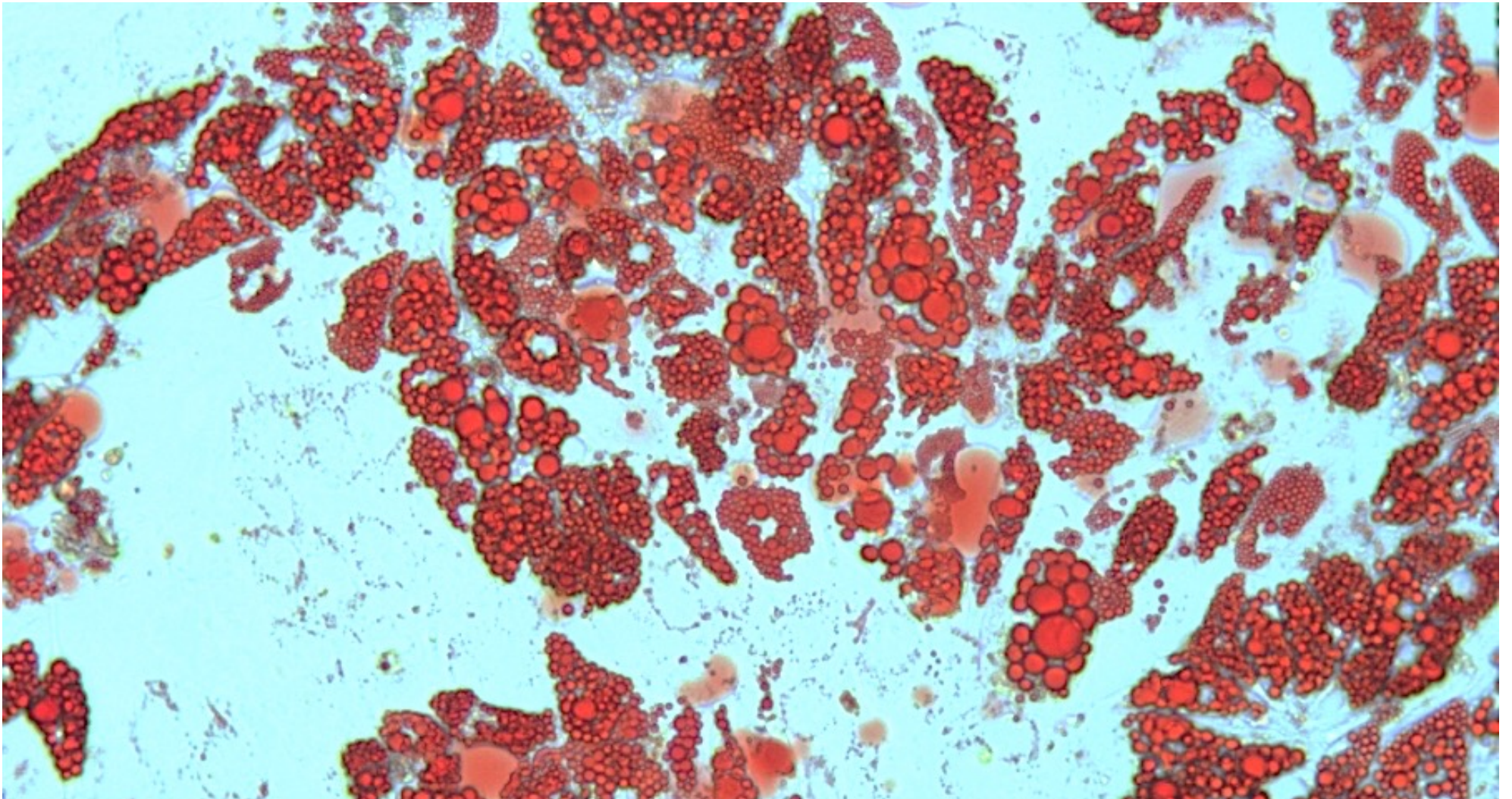
- Dermis, conjunctival stroma, etc



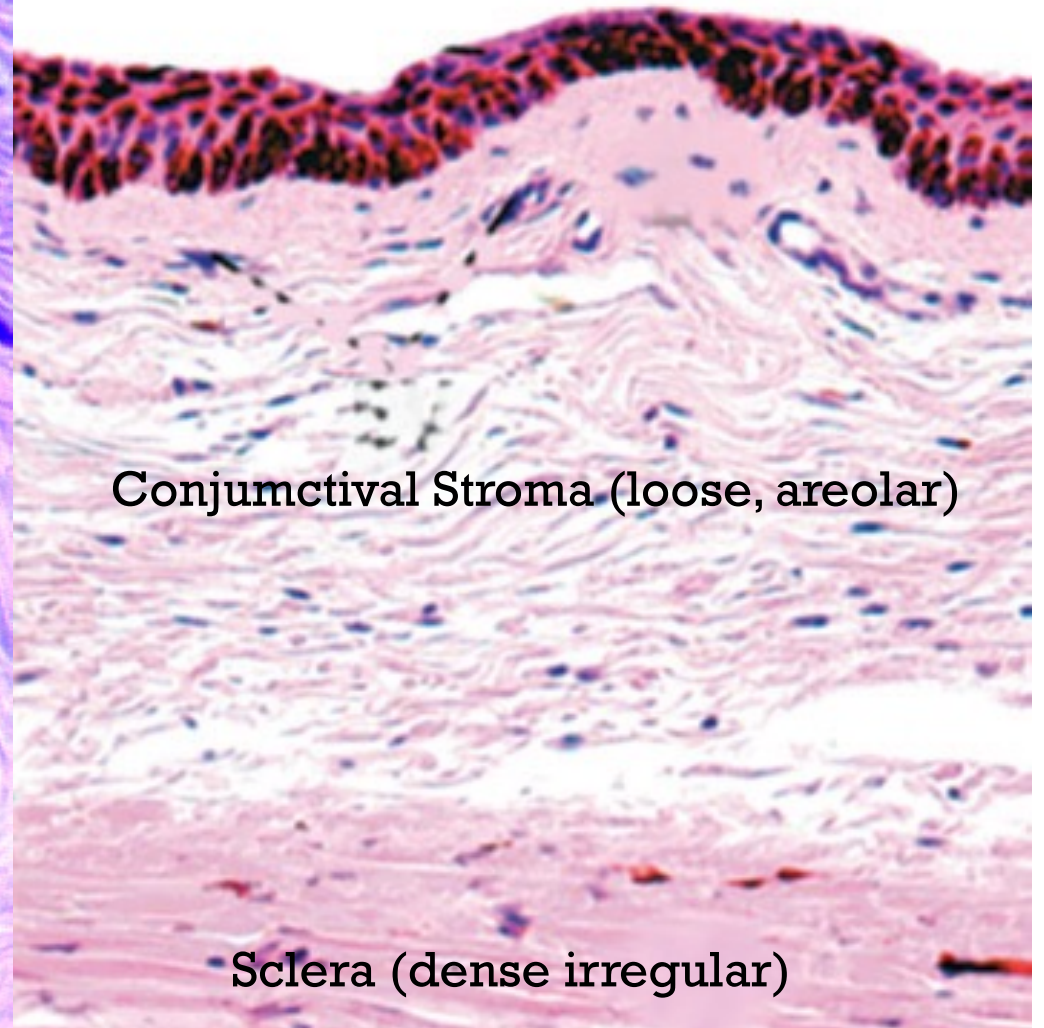
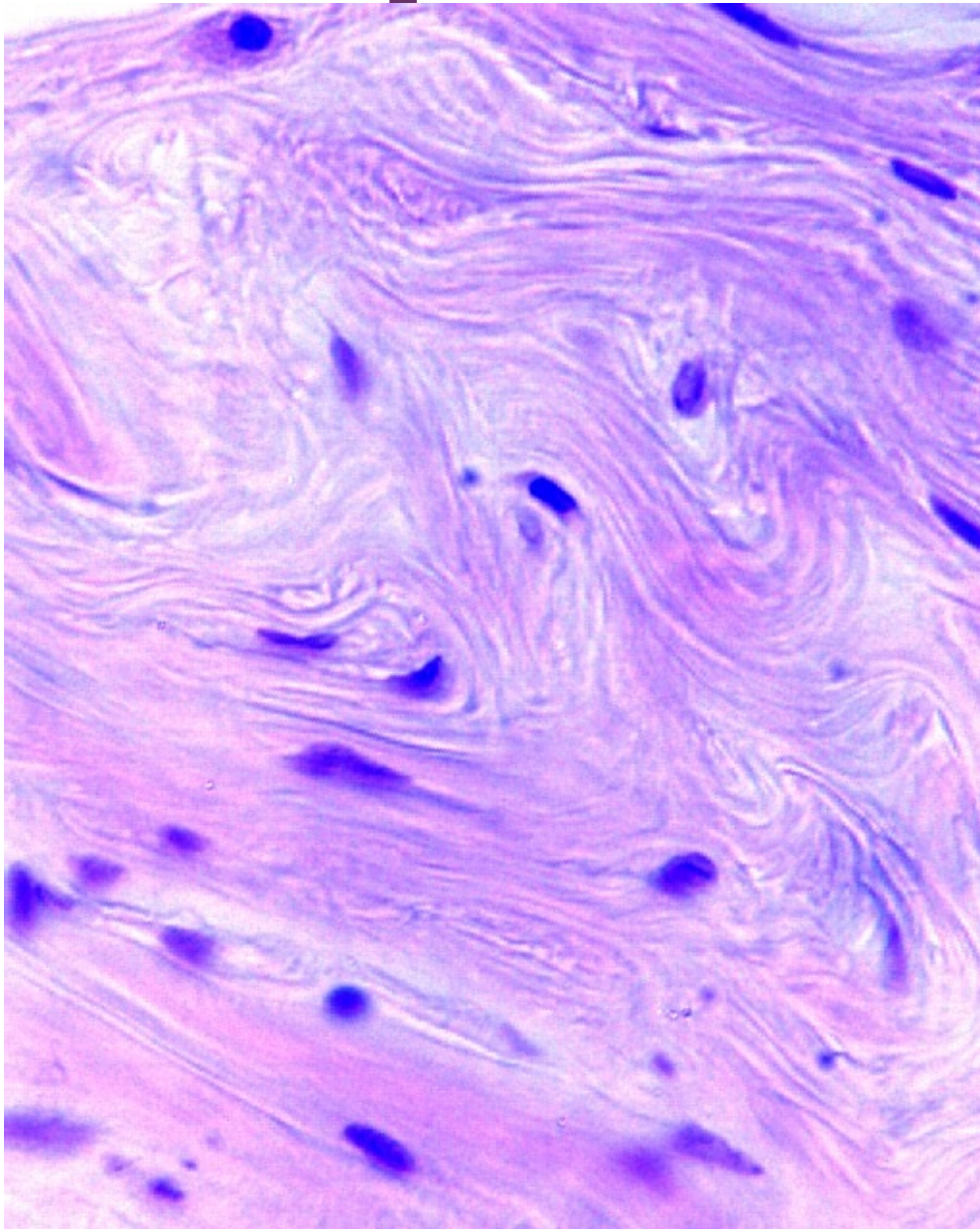
+ Adipose Connective Tissue



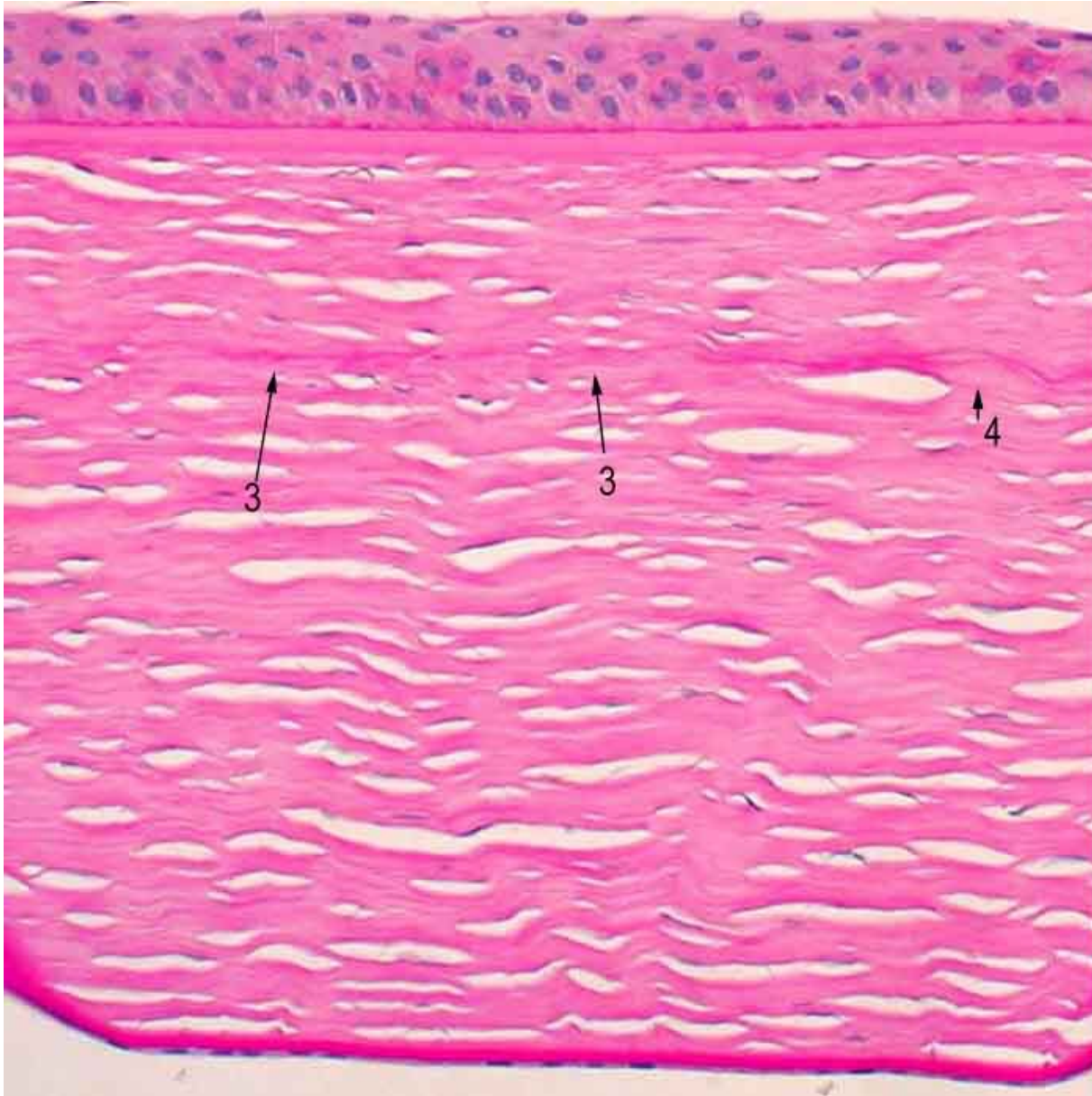
+ Frozen section of adipocytes
stained with Oil Red O



+ Dense Irregular – sclera a good example



+ Dense regular – corneal stroma is quintessential example



Elastic Connective Tissue – high elastic fiber content either alone (Bruch's membrane) or interwoven with smooth muscle (aorta).

